GTFS4EV: A Python framework for electric bus planning using GTFS data

15 december 2025

Summary

Planning the electrification of public bus fleets is a complex text, that requires modeling multiple interacting components, including electric vehicle characteristics, travel patterns, available charging infrastructure, and context-specific operational constraints and objectives. However, this typically requires detailed operational data on bus movements, which are often unavailable or incomplete in many regions, creating a major barrier to public transport electrification.

To address this gap, we introduce GTFS4EV, an open-source tool that leverages the widely available General Transit Feed Specification (GTFS) data to explore different electrification pathways for public bus fleets. GTFS4EV enables users to simulate mobility patterns, estimate energy demand, and assess the feasibility of various electrification scenarios, while providing insights into key technical requirements such as battery sizing and charging infrastructure needs. It supports the integration of custom charging strategies, allowing flexible exploration of context-specific scenarios. This makes it a powerful decision-support tool for identifying effective and tailored electrification strategies.

The tool is implemented as a modular Python library and also provides a command-line interface for users without programming experience. It enables rapid, data-driven exploration of realistic electrification scenarios, helping bus operators and urban planners assess the requirements and benefits of different fleet electrification strategies.

Statement of need

The electrification of public transport systems is a central pillar of the transition to sustainable transport. Compared to conventional diesel buses, electric buses offer reductions in greenhouse gas emissions, air pollution, noise, and operating costs (Borén 2019; Holland et al. 2020; Ghotge et al. 2025). However, planning for bus electrification remains a complex task. It requires jointly considering bus operations, charging strategies, investments in charging infrastructure and electric buses (Shyam Sundar Govinda Raja Perumal et al. 2021; Alyson da Luz Pereira Rodrigues and Seixas 2021). In some cases, bus rescheduling may also be considered (Behnia et al. 2024). These factors, in turn, shape the broader benefits and challenges for electrification, notably the potential impact on local electricity grids (Alyson da Luz Pereira Rodrigues and Seixas 2021).

In this context, a number of modelling tools and studies have been developed to support the planning process. Many existing tools - often commercial and proprietary - provide detailed operational simulations based on vehicle-level datasets such as high-resolution GPS traces and detailed schedules (e.g., EVopt Planner (Microgrid Labs 2025), eDepotPlanner (ChargeSim 2025), eflips-X (Heide et al. 2025)). While such data-rich models are well suited for advanced planning and operational fine-tuning, they are less appropriate for exploratory assessments that require a rapid evaluation of multiple electrification scenarios. Moreover, they rely on vehicle-level data that are often unavailable, particularly in emerging economies. At the other end of the spectrum, simplified tools based on generic operational assumptions (e.g., (International Energy Agency (IEA) 2023)) enable quick first-order analyses but lack spatial analysis and the ability to capture local operational specificities. As a result, there remains a gap for open and data-light modelling approaches that balance accessibility with sufficient operational realism to deliver quantitative insights.

In parallel, the increasing availability of open public transport data has created new opportunities for modelling bus electrification using openly accessible information. In particular, the General Transit Feed Specification (GTFS) has emerged as a widely adopted open standard, providing harmonized data on routes, stops, timetables, and service frequencies. Beyond the growing number of transit agencies publishing GTFS feeds, data collection initiatives have further expanded coverage (DigitalTransport4Africa (DT4A) 2024). As a result, a broad ecosystem of open-source tools has emerged to generate, process, analyze, and visualize GTFS data (GTFS.org 2025). However, only a limited number of these tools explicitly leverage GTFS data for bus electrification planning, and these address only part of the functionalty needed to support informed decision-making:

- RouteZero (Hendriks and Sturmberg 2024) focuses on depot charging optimization using mixed-integer linear programming. While effective for depot-centric analyses, it does not support opportunity or mixed charging strategies and is limited to charging demand and power analyses, without considering broader implications such as greenhouse gas emissions, air pollution, or integration with renewable energy sources.
- gtfs2emis (Vieira et al. 2023) estimates vehicle movements and emissions from GTFS data and is primarily designed for environmental impact assessment. Hence, it does not model charging behavior or infrastructure requirements, limiting its applicability for electrification planning.

- GTFS_PowerTransNet (Zhao et al. 2023) generates coupled representations of transit networks and simplified power grid topologies using GTFS data to identify candidate charging locations. While valuable for co-planning studies, it offers limited other decision-support capabilities for electrification planning.
- Eventually, some microtraffic simulation frameworks that integrate GTFS data import (e.g., tools such as EV-Fleet-Sim (Abraham et al. 2021)) can be used to model transport operations and associated vehicle electric energy consumption. While useful to fine-tune the charging demand and, in some cases, perform rerouting studies, they do not support strategic planning of charging infrastructure or battery capacity which requires to simulate not only the energy consumption but also the charging behaviour.

GTFS4EV addresses this gap by providing an open-source, GTFS-based framework for electric bus planning. It enables data-light, scenario-based simulation of fleet electrification pathways, supporting decision-making around fleet operations, charging strategies, infrastructure requirements, PV integration potential, and broader economic and environmental implications.

Functionality

GTFS4EV is a simulation framework that combines data-driven modelling of bus fleet operations with user-defined electrification scenarios to assess the feasibility and spatio-temporal charging demand of bus fleet electrification. The framework is designed for rapid exploration of scenarios rather than detailed operational optimisation. It assumes fixed, GTFS-defined bus operations, thereby isolating electrification analyses from service planning decisions.

The core functionality of GTFS4EV spans three main dimensions, which together form also the high-level simulation workflow as illustrated in Figure 1:

1. **GTFS data pre-processing**: The input GTFS feed is validated and cleaned, and can optionally be filtered (e.g. suppression of services) or enriched (e.g. addition of extra idle times at stops or terminals). The output is a consistent GTFS dataset.
2. **Fleet operation simulation**: Using the pre-processed GTFS data, GTFS4EV simulates bus fleet operations. The number of vehicles in operation and their movements are estimated to meet the GTFS schedule, enabling the simulation of individual vehicle travel patterns throughout the service day.
3. **Scenario-based charging**: Based on user-defined electrification scenarios (i.e., available charging powers, charging strategy, and electric bus energy consumption) the model estimates the charging schedules for each individual vehicle. This enables the assessment of electrification feasibility, spatio-temporal charging demand, and associated infrastructure requirements (required number of chargers at various locations, required battery

capacities).

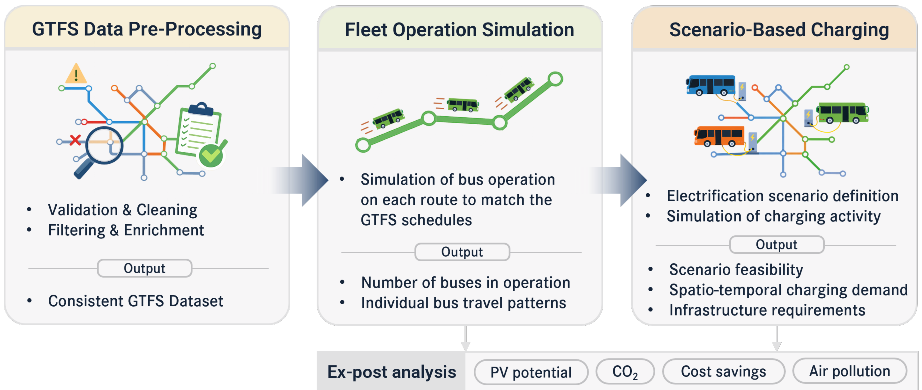

Figure 1: Overview of the three main steps of the simulation workflow

GTFS4EV also includes additional supporting features such as built-in visualization and basic analysis of GTFS feeds, and a flexible chained charging strategy logic. The framework provides default rule-based depot and opportunity charging strategies, while remaining explicitly designed to accommodate additional user-defined strategies without modifying the core simulation.

Beyond the core simulation, GTFS4EV supports a set of ex-post analyses derived from the simulated fleet operation and charging demand. In particular, it enables a preliminary assessment of PV integration potential by evaluating the temporal alignment between locally estimated PV generation and the aggregated charging demand. Additional ex-post analyses include the estimation of CO_2 emissions savings, reductions in air pollution exposure, and fuel cost savings relative to conventional diesel operations. Together, these complementary analyses broaden the decision space addressed by GTFS4EV and support the needs of the various stakeholders involved in public bus fleet electrification.

Implementation

The model is implemented as a modular Python package with an object-oriented design. Core classes correspond to the main workflow steps: GTFS data preprocessing, fleet operation simulation, and scenario-based charging demand estimation. Two additional classes support the assessment of local PV production and PV integration potential. Helper scripts provide ex-post analyses, including emissions and cost savings.

The package can be executed via a command-line interface, enabling rapid case studies and use by non-programmers. For greater flexibility and advanced usage, it can also be imported within Python scripts, allowing customized analyses and integration into other simulation workflows.

Acknowledgements

This project was supported by the HORIZON OpenMod4Africa project (Grant number 101118123), with funding from the European Union and the State Secretariat for Education, Research and Innovation (SERI) for the Swiss partners. We gratefully acknowledge the support of all OpenMod4Africa partners for their contributions and collaboration. We also express our gratitude to the developers of the PVlib toolbox (Anderson et al. 2023), which is used to support the calculation of the local PV production.

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